



KAPREKAR CONTEST – FINAL – SUB-JUNIOR

AMTI – OCTOBER, 2013.

Instructions:

1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Fill in FACE SLIP and your rough working should be in the answer book itself.
5. Maximum time allowed is THREE hours.
6. All questions carry equal marks.

1. Prove the algebraic identity $a^3 - b^3 = \left\{ \frac{a(a^3 - 2b^3)}{a^3 + b^3} \right\}^3 + \left\{ \frac{b(2a^3 - b^3)}{a^3 + b^3} \right\}^3$.
2. The ratio between a two-digit number and the sum of the digits of that number is $a : b$. If the digits in the units place is 'n' more than the digit in the tens place, prove that the number is given by $\left(\frac{9na}{11b - 2a} \right)$.
3. If $\frac{(a-b)(c-d)}{(b-c)(d-a)} = \frac{2012}{2013}$ find the value of $\frac{(a-c)(b-d)}{(a-b)(c-d)}$.
4. Q, R are the midpoints of the sides AC, AB of the isosceles triangle ABC in which $AB = AC$. The median AD is produced to E so that $DE = AD$. EQ and ER are joined to cut BC in N and M respectively. Show that $AMEN$ is a rhombus.
5. $ABCD$ is a square. The diagonals AC, BD , cut at E . From B a perpendicular is drawn to the bisector of $\angle DCA$ and it cuts AC at P and DC at Q . Prove that $DQ = 2PE$.
6. a) A two digit number is equal to six times the sum of its digits. Prove that the two digit number formed by interchanging the digits is equal to five times the sum of its digits.
b) Show that $\frac{10^{2013} + 1}{10^{2014} + 1} > \frac{10^{3013} + 1}{10^{3014} + 1}$
7. a) For any two natural numbers m, n prove that $(m^3 + n^3 + 4)$ cannot be a perfect cube.
b) A circle is divided into six sectors and the six numbers 1,0,1,0,0,0 are written clockwise, one in each sector. One can add one to the numbers in any two adjacent sectors. Is it possible to make all the numbers equal? If so after how many operations this can be achieved?
8. a) All natural numbers from 1 to 2013 are written in a row in that order. Can you insert + and - signs between them so that the value of the resulting expression is zero? If it is so how many + and - signs are inserted? Justify your answer by giving clear reasoning.
b) The natural numbers 1,2,3, are partitioned into subsets $S_1 = \{1\}$, $S_2 = \{2,3\}$, $S_3 = \{4,5,6\}$, $S_4 = \{7,8,9,10\}$ and so on. What is the greatest and least numbers in the set S_{2013} .